

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Technology (CSE)

QUIZ #2 (SET-B)**DURATION: 15 MINUTES****SUMMER SEMESTER, 2024-2025****FULL MARKS: 15**

CSE 4621: Machine Learning

Programmable calculators are not allowed.

Answer **all 5 (five)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

Student ID:

-
1. A logistic-regression classifier uses a threshold τ instead of 0.5. Which decision boundary is correct? 3
(CO1)
(PO1)
- A. $\theta^T x = \tau$
- B. $\theta^T x = \log(\tau)$
- C. $\theta^T x = \log(\tau/1 - \tau)$
- D. $\theta^T x = 1/(1 + e^{-\tau})$

Solution:

Setting $\sigma(\theta^T x) = \tau$ and solving for $\theta^T x$ gives us $\theta^T x = \log(\tau)$

2. Why is the bias term often excluded from regularization? 3
(CO1)
(PO1)
- A. The bias has no gradient
- B. Adding the bias makes the loss function non-convex
- C. The gradient of the bias is 1 as the features are normalized before training
- D. The bias only shifts the decision boundary**

Solution:

Feature weights determine sensitivity to input variables, while the bias primarily shifts the boundary. It is therefore commonly left unpenalized.

3. Why can sigmoid activations contribute to vanishing gradients in deep networks? 3
(CO1)
(PO1)
- A. The sigmoid derivative is always greater than one
- B. Its derivative is at most 0.5, so repeated multiplication can shrink gradients
- C. Sigmoid has no derivative at zero
- D. None of the above**

Solution:

Its derivative is at most 0.25. Backpropagation multiplies many derivatives. Repeated factors no larger than 0.25 can make the gradient extremely small in early layers.

4. Why can momentum accelerate training in a long, narrow loss valley? 3
- A. It removes all stochasticity from the gradient (CO1)
- B. It accumulates velocity in consistent directions** (PO1)
- C. It guarantees the global minimum
- D. The loss to become convex if sigmoid is used alongside

Solution:

Momentum reinforces motion along directions where gradients agree over time and reduces zig-zagging across steep directions.

5. Logistic regression and conventional neural-network classifiers typically assume that each input instance belongs to exactly one class. How would you design a classification system for cases in which a single instance may belong to multiple classes simultaneously? 3
- (CO2)
- (PO1)

Solution:

Use a multi-label classification model with one sigmoid output neuron per class, exactly like regular neural networks. Train the model using binary cross-entropy loss for each output, and assign every class whose predicted probability exceeds a chosen threshold.

Rubric:

- 1 mark for the correct formulation of ideas.
- 2 mark for a valid explanation.